

Cairo Air Quality vs. Respiratory Health

Analysis & Insights

Project Documentation

Track: Data Analysis Specialist

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Abstract

This project presents a comprehensive, multi-tool analytical investigation into the relationship between ambient air quality and respiratory health outcomes in Cairo, Egypt. Leveraging three integrated datasets — global PM2.5 concentration records spanning 133 countries, Egypt-specific disease burden data from the IHME Global Burden of Disease 2023 study, and macroeconomic indicators from the World Bank — the analysis delivers an end-to-end data pipeline encompassing data cleaning, Star Schema modeling, descriptive and inferential statistics, and interactive visualization across Microsoft Excel, SQL, Python, Power BI, and Tableau.

The most critical finding of this study is a sustained, statistically significant association between elevated PM2.5 exposure levels and rising DALY (Disability-Adjusted Life Years) burden attributable to chronic respiratory diseases in Cairo. Despite a partial improvement in measured air quality over the observation window, respiratory-related mortality and DALYs — particularly from Chronic Obstructive Pulmonary Disease (COPD) and lower respiratory infections — remained disproportionately high among males and the elderly, underscoring that current pollution reduction efforts are insufficient to translate into near-term measurable health gains. This finding challenges the assumption that incremental air quality improvement yields proportional health benefit, and calls for accelerated, targeted public health interventions.

Chapter 1

Introduction

1.1 Project Overview

This project presents a comprehensive multi-tool analysis of the relationship between Cairo's air quality and respiratory health outcomes. Using real-world datasets sourced from WHO/IQAir global PM2.5 records, the IHME Global Burden of Disease 2023 study (Egypt-specific), and World Bank macroeconomic indicators, the team performed end-to-end data analysis across five platforms: Microsoft Excel, SQL, Python, Power BI, and Tableau — each offering a unique analytical lens on the same underlying data.

The dataset encompasses PM2.5 concentration records for 133 countries across 2010–2025, Egypt-specific respiratory mortality and DALY data disaggregated by age, sex, and cause, and national economic indicators including GDP growth and inflation. A data preprocessing phase involved merging three complementary datasets, cleaning inconsistencies, and structuring the data into a Star Schema data model consisting of one Fact table and four Dimension tables.

1.2 Motivation

Air pollution — particularly fine particulate matter (PM2.5) — is the leading environmental health risk globally, and Cairo consistently ranks among the most polluted major cities in the world. The health burden imposed by poor air quality manifests primarily through respiratory disease, creating a direct, quantifiable link between environmental policy and population health outcomes.



These figures underscore the scale and urgency of this project. By applying descriptive, diagnostic, and inferential analytics, the team aims to identify the drivers of respiratory burden, quantify the pollution-health relationship, and deliver actionable recommendations that can directly inform public health and environmental policy.

1.3 Objectives

This project is designed to achieve the following objectives:

Objective	Description
Build a Robust Data Model	Design and implement a Star Schema (Fact + 4 Dimension tables) to enable efficient multi-dimensional analysis.
Perform Descriptive Analysis	Summarize key metrics including PM2.5 trends, mortality rates, average charges, and burden breakdown across demographic segments.
Identify Pollution Drivers	Determine which air quality indicators, demographic groups, age cohorts, and respiratory conditions are most strongly associated with health outcomes.
Apply Inferential Statistics	Validate findings using T-Tests, Chi-Square tests, ANOVA, and regression analysis to confirm that observed patterns are statistically significant.
Deliver Visual Insights	Create interactive dashboards in Excel, Power BI, and Tableau that communicate findings to both technical and non-technical stakeholders.
Provide Actionable Recommendations	Translate analytical findings into concrete public health strategies targeting emissions reduction, healthcare capacity, and population-level interventions.

1.4 Challenges

The team encountered and resolved several challenges throughout the project lifecycle:

- **Data Integration:** Merging three datasets from different sources required careful alignment of temporal coverage, country-level identifiers, and handling of missing values across PM2.5, health burden, and economic indicators.
- **Data Model Design:** Determining the optimal Star Schema structure involved critical decisions around which attributes belong in the Fact table versus Dimension tables — particularly for geographic and demographic granularity.
- **Relationship Cardinality:** Excel Power Pivot enforced 1-to-many constraints that required rethinking the geographic dimension approach, leading to a deliberate architectural decision around country-year composite keys.
- **Statistical Interpretation:** Translating p-values and regression coefficients from correlation analysis, T-Tests, and ANOVA into actionable public health language required bridging statistical rigor with policy relevance.
- **Cross-Tool Consistency:** Ensuring that insights derived from Excel, SQL, Python, Power BI, and Tableau remained consistent and complementary across all five platforms presented an ongoing coordination challenge.

Chapter 2

Methodology

This chapter documents the full analytical pipeline behind the project: where the data came from, how it was cleaned and restructured, how it was modeled, the statistical tests used to validate the findings, and how those findings were translated into dashboards across five different platforms. Each stage was cross-checked between at least two tools (Excel/SQL, SQL/Python) to guard against tool-specific calculation errors before any result was reported as a finding.

2.1 Data Sources

The analysis integrates three independently sourced datasets, merged into a single analytical base table on shared temporal and geographic keys. Each source contributes a different dimension of the story — environmental exposure, health outcome, and economic context — and none of the three alone would be sufficient to test the project's central hypothesis.

- **Air Quality Data (PM2.5)** — Global PM2.5 (fine particulate matter) concentration records for 133 countries, spanning 2010–2025, sourced from WHO/IQAir ambient air quality monitoring. This is the primary environmental exposure variable used throughout the study, and Egypt's national annual average is the value ultimately joined into the Star Schema via DIM_Macro.
- **Health Burden Data (IHME GBD 2023)** — Egypt-specific respiratory disease burden data drawn from the IHME Global Burden of Disease (GBD) 2023 study — the same academic source used by WHO and national health ministries for cross-country disease burden comparisons. It totals 77,724 individual health records, disaggregated by sex, specific age, and broader age category, across four tracked respiratory causes: Asthma, Chronic Obstructive Pulmonary Disease (COPD) / Chronic respiratory diseases, and Lower Respiratory Infections. Each record reports one of two outcome measures — Deaths or DALYs (Disability-Adjusted Life Years, a composite metric combining years of life lost to premature death and years lived with disability).
- **Macroeconomic Data (World Bank)** — National macroeconomic indicators from the World Bank, including annual GDP growth (%), inflation rate (%), and total population. These were included specifically to test the hypothesis that economic conditions — not just pollution — could be driving fluctuations in respiratory health outcomes.

Temporal Coverage and Alignment

The health burden records span 2010–2023 (14 years), while the macroeconomic and air quality series extend through 2025. Because the Star Schema join is an inner join on year, macro years without a corresponding health record (2024–2025) are naturally excluded from the combined analytical dataset — this is expected behavior, not a data-loss defect, and it means the effective

study window for any analysis that spans all three sources is 2010–2023. The 2024–2025 macroeconomic and PM2.5 records were retained separately and used as the input horizon for the forecasting model described in Section 2.4.

After the Star Schema join described in Section 2.3, the flattened master analytical table used for statistics and dashboarding contains 4,312 rows and 19 columns — one row per unique combination of year, cause, measure, sex, and age category.

2.2 Data Cleaning and Transformation

Before modeling, all four source tables were passed through a standardized cleaning and transformation pipeline, implemented consistently across Excel, SQL, and Python so that every downstream tool operates on an identical definition of each field.

2.2.1 Structural and Schema Cleaning

- **Column Name Standardization** — Column names across all four raw CSVs were normalized to lowercase, snake_case identifiers (e.g. "PM2.5" → pm2_5, "GDP Growth (%)" → gdp_growth), removing embedded spaces and percent signs from column headers to prevent join mismatches and broken references between tools.
- **Percentage Field Parsing** — GDP growth and inflation were originally stored as text fields with a trailing “%” character. These were stripped of the percent sign and cast to numeric decimals; a conditional check (mean < 1.0) was then used to detect whether a field had been left in fractional form (e.g. 0.039) versus whole-percentage form (e.g. 3.9), and fractional values were multiplied by 100 so every tool reports the same percentage scale.
- **Key Type Coercion** — Surrogate keys (demo_key, macro_key, cause_key) were converted to consistent integer types across all tables to guarantee clean joins, and a stable surrogate primary key, H_ID, was reverse-engineered for FACT_Health where one did not already exist in the raw export.

2.2.2 Data Quality Audit

A full structural audit of FACT_Health was run before any statistical work began, checking for nulls, duplicate rows, and logically invalid values (e.g. a negative death count):

Check	Result	Interpretation
Null values	0	No missing health-burden values across all 4,312 joined records.
Duplicate rows	0	No repeated year × cause × demographic × measure combinations.
Negative values (val < 0)	0	No logically invalid (negative) health burden counts.

Check	Result	Interpretation
Annual record count	Uniform across 2010–2023	Confirms the demographic stratification (Age × Sex × Cause) is fully balanced across every study year, so no year is over- or under-represented in later statistical tests.

This audit matters methodologically: uniform, complete coverage across years and demographic groups is a precondition for the ANOVA and regression tests in Section 2.4, since unbalanced group sizes can distort F-statistics and significance levels.

2.2.3 Feature Engineering

- **Population-Adjusted Burden** — A population-adjusted metric, `burden_per_100k`, was engineered as $(\text{val} \div \text{population}) \times 100,000$. This allows fair comparison of health burden across years with different national population sizes, correcting for the fact that raw burden counts naturally rise over time as Egypt's population grows, independent of any change in underlying health risk.
- **PM2.5 Risk Tiering** — Annual PM2.5 readings were bucketed into three categorical risk tiers to support non-technical reporting in the dashboards: Moderate Risk ($\leq 35 \mu\text{g}/\text{m}^3$), Unhealthy ($35\text{--}55 \mu\text{g}/\text{m}^3$), and Hazardous Exposure ($>55 \mu\text{g}/\text{m}^3$). Egypt's observed PM2.5 range ($44\text{--}86 \mu\text{g}/\text{m}^3$, see Section 2.4.1) places the entire study period within the Unhealthy-to-Hazardous band, with no year falling into the Moderate tier.
- **COVID Period Flag** — A `COVID_Period` flag was preserved on `FACT_Health` to allow later isolation of pandemic-era years (2020–2021) from the general trend, should reporting lockdown effects on mortality patterns be required.

2.2.4 Tool-Specific Adjustments

- **Excel — Relationship Cardinality Fix** — Excel's Power Pivot data model enforces a strict 1-to-many relationship cardinality between tables. Because PM2.5 values change by year within the same country, a simple country-only key created a many-to-many relationship that Power Pivot rejected. This was resolved by restructuring the geographic dimension around a composite country-year key, which required rebuilding the relevant relationship and several downstream pivot measures.
- **SQL (MySQL) — Percentile & Encoding Fixes** — MySQL-specific rewrites were required for percentile queries: MySQL does not support the `PERCENTILE_CONT` window function available in some other SQL dialects, so quartiles (Q1, median, Q3) were computed instead using a `ROW_NUMBER()` ranking approach with position-based lookups. A byte-order-mark (BOM) character present in one of the source CSVs also had to be stripped before the file would load cleanly into MySQL.

Full ETL logic — including the load, standardization, percentage cleaning, and Star Schema join steps — is implemented in the accompanying Python notebook and mirrored in the SQL data-loading scripts, ensuring the same cleaned dataset underlies every tool used downstream (Excel, SQL, Python, Power BI, and Tableau).

2.3 Data Model

The cleaned data was structured into a Star Schema consisting of one central fact table and three supporting dimension tables. A Star Schema was chosen over a normalized (3NF) design specifically to optimize for the kind of slice-and-dice, multi-dimensional analysis the dashboards required — grouping and filtering by cause, demographic segment, and year with minimal joins.

Table	Type	Key Fields	Purpose
FACT_Health	Fact	H_ID (PK), demo_key, macro_key, cause_key, val, Year, COVID_Period	One row per demographic × cause × measure × year combination; stores the health outcome value (Deaths or DALYs).
DIM_Cause	Dimension	cause_key (PK), measure, cause	Identifies the respiratory condition and whether the record measures Deaths or DALYs.
DIM_Demographics	Dimension	demo_key (PK), sex, age, age_category	Describes the population segment (sex, specific age, and broader age category).
DIM_Macro	Dimension	macro_key (PK), gdp_growth, inflation, population, pm2_5, aqi_category, year	Captures the yearly environmental and macroeconomic context (PM2.5, GDP growth, inflation, population).

Foreign keys link FACT_Health to each dimension table (demo_key, macro_key, cause_key). An inner join across all four tables — implemented identically in the SQL schema (sql_fp.sql) and the Python ETL pipeline (load_and_model_data()) — produces the unified master analytical table (4,312 rows × 19 columns) used for descriptive statistics, hypothesis testing, and every dashboard built on top of the data.

The grain of FACT_Health is deliberately fine: one row represents a single year's Deaths or DALYs count for one specific cause and one specific demographic segment (e.g. "2017, Male,

Children (0–14), Asthma, Deaths"). This grain was chosen so that every downstream aggregation — by year, by sex, by age category, by cause, or any combination — can be computed with a simple GROUP BY / SUM, without needing to re-derive or double-count any underlying record.

2.4 Statistical Analysis

Statistical analysis was carried out in two layers — descriptive and inferential — and replicated independently in SQL (using window functions and manual formula derivations) and Python (scipy.stats and statsmodels) to cross-validate every reported figure before it was written up as a finding.

2.4.1 Descriptive Statistics

Descriptive statistics were computed on the flattened master table (n = 4,312 records) for the core outcome variable (val), the population-adjusted outcome (burden_per_100k), and the three environmental/macroeconomic variables (pm2_5, gdp_growth, inflation).

Statistic	val (raw burden)	burden_per_100k	PM2.5 (µg/m³)	GDP Growth (%)	Inflation (%)
Count	4,312	4,312	4,312	4,312	4,312
Mean	12,520.14	12.90	69.07	3.95	13.09
Std. Deviation	56,272.57	60.24	15.14	1.36	8.13
Minimum	0.00	0.00	44.00	1.76	5.00
25th Percentile (Q1)	129.75	0.13	50.00	2.92	9.20
Median (50th)	1,178.00	1.17	73.50	3.99	10.25
75th Percentile (Q3)	11,433.25	11.75	82.00	5.15	13.90
Maximum	737,878.00	877.30	86.00	6.60	33.90
Interquartile Range	11,303.50	11.62	32.00	2.23	4.70
Skewness	9.21	9.80	-0.55	0.13	1.60
Kurtosis	89.79	103.70	-1.29	-0.81	1.41

Three patterns stand out in this table and shape how every later test was interpreted:

- **Severe Right-Skew in Raw Burden** — The raw health-burden variable (val) is extremely right-skewed (skewness ≈ 9.2, kurtosis ≈ 89.8): the median record burden is a modest

1,178, while the mean is pulled up to roughly 12,520 by a small number of very large outlier events (the maximum single record is 737,878). This confirms that burden is concentrated in a small number of high-severity year/cause/demographic combinations rather than spread evenly, which is why the analysis relies on non-parametric-friendly tests (ANOVA on grouped sums, Welch's t-test which does not assume equal variances) rather than assuming a normal distribution outright.

- **PM2.5 Is Consistently and Severely Elevated** — PM2.5 is comparatively tight and slightly left-skewed (skewness ≈ -0.55), ranging from 44 to 86 $\mu\text{g}/\text{m}^3$ with a mean of 69.07 $\mu\text{g}/\text{m}^3$ — between 3 and nearly 6 times the WHO annual guideline of 15 $\mu\text{g}/\text{m}^3$ in every year of the study. This confirms chronic, sustained exposure rather than isolated pollution spikes.
- **Inflation Is Far More Volatile Than GDP Growth** — GDP growth is tightly distributed (std. dev. 1.36, range 1.76–6.60%) and roughly symmetric (skewness 0.13), while inflation is far more volatile (std. dev. 8.13, range 5.00–33.90%) and right-skewed (skewness 1.60) — indicating a small number of severe inflationary shock years rather than a gradually rising trend.

The same demographic and macroeconomic extremes were also computed as targeted summary breakdowns to support the dashboards directly:

Age Category	Records	Mean Burden / Record	% of Raw Volume	% of Average Burden (Normalized)
Children (0–14)	952	34,857.88	61.47%	66.50%
Seniors (65+)	1,120	7,173.63	14.88%	13.69%
Adults (25–64)	1,792	6,038.89	20.05%	11.52%
Youth (15–24)	448	4,343.67	3.60%	8.29%

The distinction between the last two columns is the key methodological point of this table: adults (25–64) contribute the largest raw volume (20.05%) simply because that age bracket spans 40 years and therefore has the most database records (1,792). Once burden is normalized to an average-per-record basis, however, children (0–14) — despite having far fewer records (952) — carry by far the highest severity, at 66.50% of normalized average burden, roughly 5.8 times the burden-per-record of working-age adults.

Macroeconomic / Environmental Metric	Mean	Min	Max	Range (Volatility)
GDP Growth (%)	3.95	1.76	6.60	4.84

Macroeconomic / Environmental Metric	Mean	Min	Max	Range (Volatility)
Inflation Rate (%)	13.09	5.00	33.90	28.90
PM2.5 Pollution ($\mu\text{g}/\text{m}^3$)	69.07	44.00	86.00	42.00

2.4.2 Inferential Statistics

Four hypothesis tests and one regression model were run to move from descriptive patterns to statistically validated conclusions. Each test's null hypothesis, result, and interpretation are summarized below; detailed derivations are available in the SQL script (manual formula-based computation) and the Python notebook (scipy.stats / statsmodels).

Test	Null Hypothesis (H_0)	Result	Conclusion
Pearson Correlation	No linear relationship between annual PM2.5 and annual respiratory deaths	$r \approx 0.93$ (Python), $r \approx 0.95$ (SQL cross-check), $p < 0.0001$	Rejected — very strong, statistically significant positive correlation
Welch's Two-Sample t-Test	No difference in mean annual health burden between males and females	$t \approx 3.46$, $p = 0.002$	Rejected — significant gender disparity (males higher)
One-Way ANOVA (all records)	Mean health burden is equal across all four age categories	$F \approx 67.43$, $p < 0.0001$	Rejected — burden varies significantly by age
One-Way ANOVA (deaths only)	Mean mortality is equal across all four age categories	$F \approx 81.57$, $p < 0.0001$	Rejected — mortality concentrated in children and seniors (78.2% of deaths combined)
OLS Multiple Regression	PM2.5, GDP growth, and inflation jointly have no effect on annual deaths	$R^2 = 0.880$, $F = 24.44$, $p < 0.0001$	Rejected overall — but PM2.5 alone is the only significant individual predictor ($p < 0.001$); GDP growth is not significant ($p = 0.339$)
Pearson Correlation	No linear relationship between GDP growth / inflation and PM2.5	$r = -0.24$ (GDP) / $r = -0.28$ (inflation), both $p > 0.05$	Not rejected — pollution is structurally driven, not tied to annual economic cycles

Two points of nuance are worth calling out explicitly. First, the Pearson correlation between PM2.5 and respiratory deaths differs slightly between the Python run ($r \approx 0.93$) and the SQL run ($r \approx 0.95$); this gap comes from a small difference in yearly aggregation logic (the Python version aggregates directly on the flattened master table, while the SQL version re-derives yearly totals from a filtered subquery), and both estimates lead to the same conclusion — a very strong, highly significant positive relationship. Second, the OLS regression result is a joint test: the model as a whole is statistically significant ($p < 0.0001$), but only PM2.5 clears the significance threshold on its own; GDP growth does not, meaning short-term economic growth cannot be relied upon to reduce mortality independent of pollution.

In addition to hypothesis testing, a two-stage machine learning forecasting pipeline (linear and ridge regression) was built in Python to project PM2.5 levels, deaths, DALYs, and population forward from 2024 through 2033. The fitted models achieved an R^2 of 0.869 for the deaths forecast and 0.957 for the DALYs forecast on historical hold-out data, giving a reasonably reliable forward-looking view to complement the historical inferential findings above.

2.5 Dashboard Design

Findings were operationalized into interactive dashboards across four platforms, each chosen to reach a different audience and analytical depth while drawing on the exact same cleaned Star Schema described in Section 2.3, so that a KPI in one tool should always reconcile with the same KPI in another.

Excel

A workbook built on a Power Pivot data model, with the four Star Schema tables loaded as linked tables and connected through the relationships described in Section 2.2.4. On top of this model sit pivot charts and a KPI summary card layout mirroring the same headline metrics used elsewhere: total records, total deaths, total DALYs, average PM2.5, times above the WHO PM2.5 limit, study period, average annual GDP growth, and the maximum single-year inflation rate. This gives project reviewers a lightweight, offline-friendly view of the same numbers surfaced in the more interactive tools.

Python (Streamlit)

A self-contained Streamlit application (app.py) rebuilds the entire ETL pipeline and statistical suite at runtime, using `@st.cache_data` to cache both the loaded/joined dataset and the computed statistical results so the app stays responsive on interaction. The app is organized into four tabs, each targeting a different use case:

-  **Executive KPI Overview** — High-level KPI cards (total burden, PM2.5 average, mortality totals) for a quick executive read.

- **👤 Demographics & Causes** — Breakdown views by age category, sex, and cause of illness, surfacing the normalized-vs-raw-volume distinction described in Section 2.4.1.
- **🔧 Statistical Lab & Economics** — Interactive versions of the ANOVA, t-test, correlation, and OLS regression results from Section 2.4.2, alongside the macroeconomic trend charts.
- **💡 Strategic Action Center** — A summary of the recommendations in Section 2.7, framed for a policy or operational audience.

All charts across the app use a consistent dark Plotly theme (`px.defaults.template = "plotly_dark"`) for visual cohesion, and the PM2.5 risk-tier categories from Section 2.2.3 are used to color-code exposure levels throughout.

Power BI

A Power BI dashboard was built on the same Star Schema to provide interactive drill-down navigation across cause, demographic segment, and year for stakeholders who prefer a filter-driven exploration experience over a fixed set of tabs.

Tableau

A Tableau dashboard was built with an emphasis on cross-tool visual storytelling — trend lines comparing PM2.5 against mortality over time, and demographic breakdown views designed for a non-technical audience such as public health policymakers.

Note: this documentation was generated directly from the project's Excel workbook, SQL script, and Python notebook; the Power BI (.pbix) and Tableau (.twbx) files could not be opened programmatically in this environment, so the two descriptions above reflect the project's stated design intent rather than a direct visual inspection of those files. It is recommended to verify the exact chart types, filters, and KPI card layout in the Power BI and Tableau workbooks against this summary before final submission, and to update this section with any tool-specific details (e.g. exact visual names, slicers used) once reviewed.

Chapter 3 Insights and Recommendations

3.1 Insights

The five insights below represent the project's core, statistically validated conclusions, each grounded directly in the tests summarized in Section 2.4.2.

1. Air Pollution Is the Dominant, Statistically Confirmed Driver of Respiratory Mortality

Annual PM2.5 concentration correlates very strongly with annual respiratory deaths ($r \approx 0.93$ – 0.95 , $p < 0.0001$ across both the SQL and Python cross-checks), and PM2.5 remains the only statistically significant individual predictor of mortality in the multiple regression model (overall $R^2 = 0.880$). Air quality — not economic growth — is the primary, evidence-backed lever for reducing respiratory deaths in Cairo.

2. Children and Seniors Carry a Disproportionate, Non-Linear Share of the Burden

Dependent age groups absorb the overwhelming majority of harm once burden is normalized per record: children (0–14) alone account for 66.50% of normalized average burden despite representing only 22.1% of database records, and children plus seniors (65+) together account for over 80% of normalized burden and 78.2% of all recorded deaths — while working-age adults (25–64), who span 40 years of life and generate the largest raw record volume, account for only about 20% of mortality.

3. Men Bear a Significantly Higher Health Burden Than Women

Males experience a statistically significant higher annual health burden than females (mean annual burden 2,060,639 vs. 1,795,563; Welch's $t \approx 3.46$, $p = 0.002$), consistent with greater outdoor and occupational exposure to airborne pollutants among the male working population.

4. Inflation Shocks Compound the Health Crisis Independently of Pollution Levels

While GDP growth and inflation show no direct statistical correlation with PM2.5 levels ($r = -0.24$ and $r = -0.28$ respectively, both $p > 0.05$), inflation itself is highly volatile — averaging 13.09% but peaking at 33.90%, a swing of nearly 29 percentage points. These inflation shocks coincide with reduced healthcare affordability, suggesting an indirect but real pathway through which economic instability can worsen respiratory outcomes even in years when air quality is unchanged or improving.

5. Measured Improvement in Air Quality Has Not Yet Translated Into Proportional Health Gains

Despite some improvement in measured air quality over the observation window, respiratory mortality and DALYs have remained disproportionately high across the study period, indicating that current pollution-reduction measures have not yet produced a proportional, near-term

reduction in respiratory harm — a lag effect that has direct implications for how quickly policy interventions should be expected to show measurable results.

3.2 Recommendations

Based on the statistical evidence in Sections 2.4 and 2.6, the following actions are recommended, ordered by the population segment or system they target. Each recommendation is tied directly back to the finding that motivates it.

- **Accelerate Emissions Reduction** — Expand PM2.5 monitoring and enforcement in high-density Cairo districts, and prioritize interventions that directly reduce ambient particulate concentration (vehicle emissions, industrial filtration, construction dust control). This is the single highest-leverage action supported by the data, since PM2.5 is the only statistically significant direct predictor of mortality identified in the regression model (Section 2.4.2).
- **Prioritize Pediatric and Geriatric Respiratory Care** — Direct incremental healthcare capacity — pediatric respiratory clinics, school-based screening programs, and geriatric respiratory support services — toward children and seniors, who together account for over 80% of normalized health burden and 78.2% of recorded deaths despite representing a much smaller share of total population than working-age adults (Section 2.6, Insight 2).
- **Target Occupational Exposure Among Male Workers** — Introduce occupational health protections, workplace air-quality standards, and voluntary screening programs for outdoor and industrial male workers, given their significantly higher measured health burden relative to women (Section 2.6, Insight 3).
- **Buffer Healthcare Affordability During Inflation Shocks** — Protect access to essential respiratory medicines (inhalers, bronchodilators) through price controls or inflation-indexed subsidies during periods of high inflation, to prevent the affordability collapse that appears to drive mortality spikes independently of pollution levels (Section 2.6, Insight 4).
- **Monitor Forecast vs. Actual Outcomes Annually** — Track the forecasted 2024–2033 trajectory of PM2.5, deaths, and DALYs produced by the project's forecasting model (R^2 of 0.869 for deaths, 0.957 for DALYs on hold-out data) against real outcomes each year, and treat material deviations from the forecast as an early signal that policy adjustments are needed before the next annual reporting cycle.
- **Institutionalize the Data Model for Future Reporting Cycles** — Maintain a single governed Star Schema (as built in this project) as the shared source of truth across Excel, SQL, Python, Power BI, and Tableau, so future reporting cycles remain consistent across tools and stakeholders, and so that any KPI reported in one tool can always be reconciled against the same KPI in another.
- **Set Realistic Timelines for Measurable Health Impact** — Because the improvement in measured air quality has not yet produced a proportional drop in mortality (Section 2.6,

Insight 5), set policy expectations around a realistic lag time between an emissions intervention and its measurable health impact, rather than expecting an immediate year-over-year drop in deaths or DALYs.